BioHack 2026 – HeatSignal NYC (Proposal)

Team Name: [Team name] **Team Members:** [Member names] **Selected Stream:** Public Health & Epidemiology

Specific research question or problem being addressed:

NYC issues a heat advisory only when the heat index hits 100°F, or 95°F two days running [2]. Does that rule warn on the days New Yorkers actually get sick from heat? Can a forecast of heat ER visits, tied to neighborhood vulnerability, warn earlier and send low-cost help where it is needed most?

Background, context & significance:

Heat is NYC's deadliest weather hazard: about 500 premature deaths a summer. Black New Yorkers die of heat stress at twice the white rate; most heat-stress deaths occur in homes without AC [1]. Daily heat ER visits track excess deaths [4]. The 95 to 100°F range matches a study of 1997 to 2006 deaths that urged re-checking with new data [5], and early heat waves are about twice as deadly [6]. NWS HeatRisk is national and climate-based, not tuned to NYC outcomes [7]. **We found: 56% of 7,619 heat ER visits (2017 to 2026) fell on days with no advisory.** It missed 61 of 158 high-burden days. At 91 to 97°F, early-season days had 35% more visits.

Proposed solution:

HeatSignal NYC turns each morning's NWS forecast into expected heat ER visits (with an 80% range) for 3 days. It adds a **Heat Watch** tier (9.35+ expected visits, about 20 days a summer) under the unchanged advisory. A Watch triggers low-cost actions in the 81 neighborhoods with Heat Vulnerability Index 4 or 5: calls to older adults living alone, texts, hospital heads-ups and longer cooling-center hours, guided by a ranked call list. **New:** a city trigger calibrated to NYC's own ER outcomes, stated in expected patients, with each action's cost matched to how certain the alert is.

Data & methodology:

Data: NYC Health Dept. (public) [3]: daily heat ER visits from syndromic surveillance of every NYC ER, and max heat index at LaGuardia, May to Sept. 2017 to 2026 (1,520 days); HVI for 198 neighborhoods. **Method:** Poisson regression on heat index (extra slope above 85/90/95°F), prior-day heat, hot streaks, early-season heat, weekend, and visits 2 days earlier. Leave-one-summer-out validation with cutoffs set on training years, ±2.5°F forecast error simulated, and a script that re-checks all 44 numbers we cite. **Results:** mean forecast matches mean actual in every forecast range (e.g., 14.3 vs. 14.8). Watch + Advisory warned on **131 of 158 bad days (83%, 95% CI 75 to 91%) vs. 97 (61%, 51 to 74%)**. A warning was in place from day one in 46 vs. 30 of 62 heat episodes, and in 82% of bad days with forecast error.

Prototype plan and future goals:

Built: an in-browser app with a 3-day planner (live NWS forecast), an out-of-sample replay of any summer (June 2024: Watch a day before the advisory), and a neighborhood map with call list. **Next:** ZIP-level forecasts with Health Dept. data, adding overnight heat and ozone, and a one-summer South Bronx pilot measuring calls made and ER visits.

Impact, implications & application:

Users: Health Dept., NYC Emergency Management, outreach groups, hospitals. Beneficiaries: South Bronx, Upper Manhattan and central Brooklyn residents (HVI-5 areas as low as 76% AC). Over 10 summers, about 1,000 more heat ER visits fell on warned days (about 100 a summer) for about 8 extra alert days a year. A false Watch costs some calls; a miss can cost a life. Any city among the 85% of US ERs reporting to CDC syndromic surveillance [8] could reuse it.

Feasibility & execution plan:

Built this weekend from public data, no servers. Roles: [Member 1] data and model; [Member 2] validation; [Member 3] web app; [Member 4] playbook and pitch. Deploying needs no new data system: the city gets these counts daily. The model is refit each fall.

Limitations:

Counts are suspected illness and miss deaths at home and people who avoid ERs. One weather station; citywide outcomes only. A plain lower heat-index cutoff gets close (129 vs. 131), so most gain comes from warning at lower temperatures; the model adds patient estimates and precision (66% vs. 61%). It underpredicts the worst early heat waves (June 25, 2025: 31 expected, 141 actual). Watch goes to outreach teams, not sirens, to limit alert fatigue.

Cite all references and tools used below*

[1] NYC DOHMH. 2026 Heat-Related Mortality Report. a816-dohbesp.nyc.gov/IndicatorPublic/data-features/heat-report [2] National Weather Service New York. Heat Advisory criteria. weather.gov/okx/extremeheat

[3] NYC DOHMH Environment & Health Data Portal data (heat-syndrome ED visits; Heat Vulnerability Index). github.com/nychealth/EHDP-data, accessed Oct 3, 2026.

[4] Mathes RW et al. PLoS One. 2017;12(9):e0184364. doi:10.1371/journal.pone.0184364 [5] Metzger KB, Ito K, Matte TD. Environ Health Perspect. 2010;118(1):80-86. doi:10.1289/ehp.0900906

[6] Anderson GB, Bell ML. Environ Health Perspect. 2011;119(2):210-218. doi:10.1289/ehp.1002313 [7] NOAA/CDC. NWS HeatRisk (2024), noaa.gov [8] CDC. About NSSP, cdc.gov/nssp

Live forecast: NWS API. Tools: Python (pandas, scikit-learn), JavaScript. Code and fact check (FACTS.md) included with the submission files.